

Lesson 3: Quantum Measurement with a One-Qubit Circuit

Lesson 3: Quantum Measurement with a One-Qubit Circuit Hadamard, measurement, histogram, and probabilistic output

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Learning Outcomes

- Create a one-qubit circuit with Qiskit.
- Explain the Hadamard gate as an equal-superposition operation.
- Explain measurement counts and why they fluctuate.
- Read a simple quantum histogram.

The Quantum Idea in One Sentence

A quantum circuit prepares a state, applies gates, and measurement turns that state into classical data.

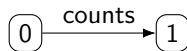
- A qubit starts in a basis state such as $|0\rangle$.
- The Hadamard gate creates an equal superposition.
- Measurement returns 0 or 1.
- Repeated shots reveal the distribution.

Build the Circuit

```
from qiskit import QuantumCircuit
from qiskit_aer import Aer
from qiskit.visualization import plot_histogram
import matplotlib.pyplot as plt

qc = QuantumCircuit(1, 1)
qc.h(0)
qc.measure(0, 0)
simulator = Aer.get_backend('aer_simulator')
job = simulator.run(qc, shots=1000)
result = job.result()
counts = result.get_counts(qc)
print(counts)
labels = list(counts.keys())
values = list(counts.values())
colors = ['blue', 'red']
plt.bar(labels, values, color=colors)
plt.xlabel("Measurement Outcome")
plt.ylabel("Count")
plt.title("Quantum Measurement Results")
plt.show()
```

Interpreting the Histogram



- The notebook reports '1': 512, '0': 488 in this run.
- That is approximately 51.2% for 1 and 48.8% for 0.
- The exact numbers can change on another run because measurement is probabilistic.

Why the Numbers Are Not Exactly 500/500

- The ideal probabilities are 50% and 50%.
- 1000 shots are finite samples from that distribution.
- Sampling variation is expected.
- More shots generally make the observed proportions closer to the theoretical probabilities.

What the Plotting Code Does

- `labels` contains the measured bit strings.
- `values` contains their counts.
- `plt.bar` creates a classical bar chart.
- The chart is a visualization of measurement data—not the quantum state itself.

Prerequisite Bridge

- Know that a classical bit has two basis values: 0 and 1.
- Know that probability describes repeated outcomes.
- Recognize that Qiskit uses Python syntax, but the objects represent quantum circuits.

Hands-on Activity

- Remove the Hadamard gate and rerun.
- Predict the counts before execution.
- Restore the Hadamard gate and compare.
- Try 100, 1000 and 10,000 shots.

Lesson Summary

- A Hadamard gate creates an equal superposition from $|0\rangle$.
- Measurement gives classical results.
- Counts are empirical observations of probabilities.
- Qiskit plus Aer lets beginners experiment without hardware.

A one-qubit circuit already introduces state preparation, gates, measurement and sampling.

Quick Check

- ① What does `qc.h(0)` do?
- ② Why is `shots=1000` useful?
- ③ What does `get_counts()` return?
- ④ Why can two runs give different counts?